



QINGDAO LONGXIN WIND POWER PROJECT PHASE I

JUNO.

DOCUMENT PREPARED BY JUNO CARBON INVESTMENT & ENVIRONMENTAL
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Project Title	Qingdao Longxin Wind Power Project Phase I
Version	02
Date of Issue	29-08-2022
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Qingdao Longxin Wind Power Project Phase I (hereinafter referred to as the Project) developed by China Resources Wind Power (Qingdao) Co., Ltd., involves construction and operation of a wind power project that is sited on Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The Project was commenced on 08/11/2011. The Project has been registered as a VCS project on 19/04/2012 and registered as a CDM project on 28/09/2012 (UNFCCC registration reference number: 6697).

The purpose of the Project is to install 24 wind turbines with capacity of 2000KW each and 1 wind turbine with capacity of 1800kW to generate clean renewable electricity. The electricity generated by the Project is supplied to the North China Power Grid (NCPG), displacing part of electricity generated by NCPG which is dominated by fossil fuel fired power plants. According to the methodology ACM0002 (version 20.0), the baseline emissions are the electricity produced by the project multiplied by the emission factor of NCPG.

Wind turbines and their corresponding generators employed by the Project were provided by Vestas Wind Technology (China) Co., Ltd.

Relevant dates for the project activity is as below: Construction start date: 26/05/2011
Commissioning start date: 08/11/2011

The estimated annual net electricity generation and average annual emission reductions of the proposed project are 99,026MWh and 81,885tCO₂e, respectively.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1. Energy industries (renewable/non-renewable sources);

Project type: Grid-connected Wind power generation project;

The project is not a grouped project, which is defined by the VCS standard v4.3 as a number of projects and their related methodologies included in a single VCS PD at the time of the validation. The project is a single project in a single location.

1.3 Project Eligibility

The project is a wind power project, which reduces CO₂ by replacing electricity from fossil fuel power plants.

The project is eligible under the scope of the VCS program in accordance with VCS standard v3.3 when the project was accepted by the VCS board (No. #969), And the eligibility is still valid during the second crediting period.

1.4 Project Design

The project was designed as a single installation activity, not under a group project.

1.5 Project Proponent

Organization name	China Resources Wind Power (Qingdao) Co., Ltd.
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1.6 Other Entities Involved in the Project

Organization name	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.
Role in the project	VCU Buyer
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1.7 Ownership

The project owner is China Resources Wind Power (Qingdao) Co., Ltd., who has the full ownership of the project. The approval of Environmental Impact Assessment (EIA), Feasibility Study Report (FSR), and Letter of Approval for the Project as a CDM Project issued by China National

Development and Reform Commission, and the business license of the project owner are evidences for legislative right. Besides, the equipment purchasing contract and the power purchasing agreement are the evidences for the ownership of the plant, equipment and power generated.

1.8 Project Start Date

The project start date is 08/11/2011 when the electricity was generated and supplied to the North China Power Grid (NCPG).

1.9 Project Crediting Period

The project crediting period is ten years, twice renewable for a total of 30 years. However, the project is also registered as a CDM project (CDM ID: 6697) and the crediting period under CDM is 21 years (7*3 renewable), it is not eligible for VCU issuance beyond the end of those 21 years. The project's first crediting period is from 08/11/2011 to 07/11/2021. Now the project's second crediting period from 08/11/2021 to 07/11/2031 is going to be renewed.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
8/11/2021-7/11/2022	81,885
8/11/2022-7/11/2023	81,885
8/11/2023-7/11/2024	81,885
8/11/2024-7/11/2025	81,885
8/11/2025-7/11/2026	81,885
8/11/2026-7/11/2027	81,885
8/11/2027-7/11/2028	81,885
8/11/2028-7/11/2029	81,885
8/11/2029-7/11/2030	81,885

8/11/2030-7/11/2031	81,885
Total estimated ERs	818,850
Total number of crediting years	10
Average annual ERs	81,885

1.11 Description of the Project Activity

Qingdao Longxin Wind Power Project Phase I is to utilize wind resources for electricity generation through the construction of a wind farm with a total capacity of 49.8MW in Liyuan Town and Dianzi town, Pingdu City, Shandong Province, P. R. China. The electricity generated from the project is sold to North China Power Grid (NCPG). The project achieves obvious greenhouse gas (GHG) emission reductions through the displacement of electricity delivered by North China Power Grid which is a fossil-fuel dominated grid.

The purpose of the proposed project is to generate zero-emission wind power and deliver it to North China Power Grid. For the project,

(a) Prior to the start of implementation of the project activity, there is no power generation unit at the site of the proposed project, and the electricity was supplied by the North China Power Grid which was dominated by fossil fuel-fired power plants.

(b) The project scenario is the implementation of the project, 24 wind turbines with capacity of 2000 kW each and 1 wind turbine with capacity of 1800 kW with a total capacity of 49.8MW supply an average annual generation of 99,026 MWh to North China Power Grid and replace the same amount of electricity generated by fossil fuel-fired power plants connected to North China Power Grid. According to ACM0002 applied, the project is a zero-emission electricity generating activity.

(c) The baseline scenario of the project is the electricity supply of equal amount as the project from the North China Power Grid. The baseline scenario of the project is the same as the scenario prior to the start of the implementation of the project activity.

The project involves 24 wind turbines with capacity of 2000 kW each and 1 wind turbines with capacity of 1800 kW which amount to a total installed capacity of 49.8MW. The project is constructed and operated by China Resources Wind Power (Qingdao) Co., Ltd. The estimated annual net electricity generation and average annual emission reductions of the proposed project are 99,026MWh and 81,885tCO₂e, respectively.

Technologies applied on the project activity

The project consists of one site only and the implementation is not phased. This project installs 24 wind turbines with capacity of 2000 kW each and 1 wind turbines with capacity of 1800 kW

which amount to a total installed capacity of 49.8MW. The estimated power output is 99,026MWh and the load factor is 22.70%.

The main technical specifications of the wind turbine are provided in the following table.

Item	Unit	Index	Index
Model		V90-2.0MW	V90-1.8MW
Rated capacity	kW	2000	1800
Number of blades		3	3
Rotor diameter	m	90	90
Rated wind speed	m/s	13	12
Shutoff speed	m/s	2.5	3.5
Height of hub	m	80	80
Rated	V	690	690

An 110kV substation is built at the project site, which connects the project to the Tangtian 220kV Substation which is about 6.7 km away from the project and belongs to the North China Power Grid. Then the power produced in the wind farm can be transmitted to the North China Power Grid.

1.12 Project Location

The project is located at Liyuan Town and Dianzi Town, Pingdu City, Shandong Province, P.R. China. The nearest Liyuan Town is about 10 kilometers from the project site. The geographical coordinates of the project is from 119.8827°E to 119.9503°E, and from 36.8157°N to 37.8915°N (The central GPS coordinate is 119.91125°E, 36.8568°N). Figure A1 and A2 show the geographical location of the project.



Figure 1. The project on the map of P. R. China



Figure 2. The project on the map of Pingdu City, Shandong Province, P. R. China

1.13 Conditions Prior to Project Initiation

The project is to utilize wind power source to generate electricity and deliver it to NCPG. According to the methodology ACM0002 (Version 20.0), no emission sources and greenhouse gases (GHGs) are involved in the project activity since it is the installation of a new grid-connected renewable energy project.

For the Project, the scenario existing prior to the start of the implementation of the project activity is the electricity estimated to be generated was obtained from NCPG. The baseline scenario is the same as the scenario existing prior to the start of implementation of the project activity.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is a renewable energy project, the Environmental Impact Assessment (EIA) report was approved by Qingdao Environment Protection Bureau on 12/10/2010, and the Feasibility Study Report of the project was approved by National Development and Reform Committee on 02/03/2011. The project has received the Notification Form from China DNA on 22/02/2012. These approvals demonstrate that the project meets the requirement of national laws and regulations, including Renewable Energy Law of the People's Republic of China and Environmental Protection Regulation for Wind Power Project, etc.

The project is also in line with current laws and regulations, such as Safe Production Law of the People's Republic of China, Regulation on the Quality Management of Construction Projects, etc.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project was registered as a Clean Development Mechanism (CDM) project in UNFCCC on 28-Sep-2012 (UNFCCC Ref. 6697), with the renewable crediting period of 3*7 years under the CDM GHG Program. The first 7-year renewable crediting period under CDM is from 28/09/2012 to 27/09/2019. Please refer to the following link for details.

<https://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view>

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by other GHG Programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project has been registered as a CDM project in UNFCCC on 28/09/2012 (UNFCCC Ref. 6697), for which a renewable crediting period of 3*7 years will be used under the CDM GHG Program. The first 7-year renewable crediting period under CDM is from 28/09/2012 to 27/09/2019. And there are no credits have been issued under CDM mechanism.

The project has not been counted or used under GS project or under any other voluntary carbon crediting scheme. In the future, the emission reductions that apply for issuance under VCS will

not be issued under CDM or GS project or under any other voluntary carbon crediting scheme. The project does not involve in ETS or other binding limits.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates.

1.17 Sustainable Development Contributions

The Project will contribute to sustainable development in the following ways:

1.Ensure access to affordable, reliable, sustainable and modern energy for all

The project generates electricity from the sustainable and renewable wind source and contributes to increase the share of renewables in the global energy mix. The project provides 99,026 MWh renewable energy generation annually during the lifetime.

2.Promote, inclusive and sustainable economic growth, full and productive employment and decent work for all

The conducting of the project will offer 20 job opportunities for local people during the operational period. The region can achieve economic growth through the operation of the project. Furthermore, the project will contribute to local government with more tax revenues and poverty eradication.

3.Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

The project provides clean and renewable energy by adopting the environmentally wind power generation technology. The project provides 99,026 MWh renewable energy generation annually during the lifetime.

4.Take urgent action to combat climate change and its impacts

The proposed project is to replace grid-connected fossil fuel-fired power plants in the North China Power Grid, and thus reduce fossil fuel consumption and avoid pollutants emission, such as sulfur dioxide and dust, brought by fossil fuel combustion. The project will achieve a GHG emission reduction of 81,885 tCO₂e/yr during second crediting period. Therefore, the proposed project has obvious environmental benefit.

1.18 Additional Information Relevant to the Project

Leakage Management

NA.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

NA.

2 SAFEGUARDS

2.1 No Net Harm

There are no negative environmental and/or socio-economic impacts due to the project. During the second crediting period, the project as a clean renewable energy project can reduce greenhouse gas emissions and the environmental pollution caused by fossil fuels consumption. Meanwhile, the implementation of the project will improve local socio-economic development through creating career opportunities and paying taxes.

2.2 Local Stakeholder Consultation

Local Stakeholder Consultation during the project design stage: A stakeholders meeting was held by the project owner on 20-Mar-2011 in order to collect the comments of the stakeholders in local area towards the construction and operation of this project. Before the meeting, the project owner informed the local government about the time and content of the meeting and asked the government to invite representatives of the stakeholders to attend this meeting. Totally 50 stakeholder representatives participated the meeting, respectively from the local government, surrounding villages (Zhujiaying Village, Shuangmiao Village and Zhuangjiaying Village which belong to Liyuan Town). The representatives were selected from different ages, different occupations and different education levels. At this meeting, the project owner described this project activity to allow the local stakeholders to understand the project activity Questionnaire survey. During the survey of stakeholders, the comments from the relevant stakeholders were collected. 50 copies questionnaires were distributed and 49 pieces of reply were received. The recovery rate is 98%.

The summary of questionnaire survey is listed as the following:

- 37 (76%) of them think the current environment quality is good, 10 (20%) think it is normal and 2 (4%) of them are unsure;

- 38 (79%) of them think there will not be any negative impacts on their everyday life, 11 (27%) of them are not sure;
- 36 (73%) of them think the proposed project will help to improve the local economy, 13(27%) of them are not sure;
- 20 (41%) of them think the proposed project doesn't impact on local sound/noise environment, 5 (10%) of them think it will, 27 (49%) of them are unsure;
- Regarding the construction and operation of the propose project, 27 (55%) of them are most concerned with electromagnetic interference, 19 (40%) of them are most concerned with the noise level and 3 (6%) of them are most concerned with wastewater;
- 45 (92%) of them support the implementation of the proposed project and 4 (8%) of them don't care.

The summary of local stakeholders meeting:

The local community possesses basically positive comments on the effects of the proposed project. The interviewees considered that local social, economic and environmental development would benefit from the proposed project. The response was overall supportive to the project implementation.

2.3 Environmental Impact

The environmental impact assessment for this project was completed in September 2010 and approved by Qingdao Environmental Protection Bureau on 12th October, 2010. The approval number is Qinghuanshen[2010] 209. A summary of the report is illustrated as below:

Ambient air

The impact on ambient air quality of the project is mainly from dust during construction stage, by sprinkling water frequently and timely clearing can reduce the dust pollution. When the project is in operational period, there will be no air pollutions. In conclusion, the proposed project will not pose any threat on the quality of ambient air.

Impact from noise

There is some noise during the operation of wind turbines. The equipments and techniques with lower noise will be chosen to apply. Improvement on construction process and strengthening of equipment maintenance is emphasized. The Standard of noise limits of construction site (GB12523-90) should be fulfilled during the construction period and Emission standard for industrial enterprise noise at boundary (GB12348-2008) Level II noise standard would be fulfilled during operational period. Consequently, the noise of operation has little impact to the

surrounding environment. Hence, the noise will not impact the work and daily life of local residents.

Electromagnetic impact

The operation of the wind farm will generate electromagnetic pollution, whereas the pollution is slight. In addition, the wind turbines are very far from local residents and village. Therefore, the electronic magnetic pollution to the surrounding environment is insignificant.

Impact from Solid waste

Solid wastes generated from the project activity are excavated earth material and municipal solid waste. Part of the excavated earth material will be backfilled, and the rest will be used for land levelling and road construction near the project site. The municipal solid waste will be collected and treated together with the waste from local residents. As the report indicates, solid waste is handled properly.

Impact from Wastewater

Wastewater is mainly domestic wastewater. Wastewater quantity is fairly small and treatment methods will be applied for on-site primary treatment, and then the wastewater will be treated together with the local wastewater. Oxidation pond should be built on the site, through which the discharging wastewater would be fulfilled the Reuse of recycling water for urban and Water quality standard for urban miscellaneous water consumption (GB/T 18920-2002) will be used as virescence water. Therefore, the impact of wastewater is limited and mitigated.

The land where the project located belongs to wasteland and no immigration is involved in the project. Land compensation is carried out according the relative national regulations.

No migrating birds have been found in the project field. Therefore, the project is not located on the passage of migrating birds, and the project construction will not influence the migration of birds.

2.4 Public Comments

There are no negative comments received during the project construction and operation stage.

And in the public comment period under CDM program, there is no comments received.

2.5 AFOLU-Specific Safeguards

This is not an AFOLU project. Therefore, this section is not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The approved methodology applied in the proposed project activity is ACM0002 (version 20.0) – “Grid-connected electricity generation from renewable sources”.

Reference:

<https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG>

The applied tools:

“Tool for the demonstration and assessment of additionality (version 07.0.0)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

“Combined tool to identify the baseline scenario and demonstrate additionality (version 07.0)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf>

“Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion (version 03.0)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf>

“Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0) “

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf>

“Tool to calculate the emission factor for an electricity system (version 07.0)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

“Tool to determine the remaining lifetime of equipment (Version 01)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-10-v1.pdf>

“Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period (version 03.0.1)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-11-v3.0.1.pdf>

“Positive lists of technologies (version 04.0)”

Reference:

<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-32-v4.0.pdf>

3.2 Applicability of Methodology

The approved methodology ACM0002 (version 20.0) is applicable to the project activity and the project meets the applicability of the applied methodology as follows:

No.	Applicability criteria	Justification
1	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s)	Applicable. The project is a newly built wind power generation project. It includes activities that install a Greenfield power plant.
2	The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of	Applicable. The project is a newly built wind power generation project, and the project does not include capacity additions, retrofits, rehabilitations or replacements.

No.	Applicability criteria	Justification
	baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	
3	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	Irrelevant. The project does not include hydro power plants.
4	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or	Irrelevant. The project does not include hydro power plants.

No.	Applicability criteria	Justification
	(b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
5	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	Applicable. The project is a newly built wind power generation project and does not involve switching from fossil fuels to renewable energy sources at the site of the project activity. And the project does not include biomass fired power plants/units.
6	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Irrelevant. The project is a newly built wind power generation project, and does not include retrofits, rehabilitations, replacements, or capacity additions.

In addition, the project meets the applicability conditions of the applied tools applied in the section 3.1 of this VCS PD as follows:

Tool	Applicability criteria	Justification
Tool to calculate the emission factor for an electricity system (Version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable This project replaces grid power supply and uses this tool to calculate the values of OM, BM and CM of this project.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Applicable The emission factor for this project electricity system was calculated for grid power plants.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country. Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Irrelevant. There is no part of the power system of this project is located in an Annex I country. And the project does not involve biofuels.
Baseline, project and/or leakage emissions from	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:	Applicable The electricity consumption lies to Scenario A:

Tool	Applicability criteria	Justification
electricity consumption and monitoring of electricity generation (version 03.0)	(a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	Electricity consumption from the grid. The electricity is purchased from the grid only, and no captive power plant is installed at the site of electricity consumption.
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable. The project scenario lies to Scenario I: Electricity is supplied to the grid.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO2 emissions.	There is no captive renewable power generation technologies installed to provide electricity in the project activity, in

Tool	Applicability criteria	Justification
		the baseline scenario or to sources of leakage.
Assessment of the validity of the original/current baseline and update the baseline at the renewal of the crediting period (Version 03.1)	This tool provides a stepwise procedure to assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period.	Applicable The baseline in this PD is at the renewal of the crediting period. The baseline is assessed by the procedure of this tool.

3.3 Project Boundary

According to the methodology ACM0002, the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the CDM project power plant is connected to.

According to 2019 Baseline Emission Factors for Regional Power Grids in China issued by the Ministry of Ecology and Environment of China (China MEE), NCPG consists of Beijing Power Grid, Tianjin Power Grid, Hebei Power Grid, Shanxi Power Grid, Shandong Power Grid and Inner Mongolia Power Grid.

The GHG emission sources included in or excluded from the project boundary are as follows:

Source		Gas	Included?	Justification/Explanation
Baseline	Fossil fuel-fired power plants in the NCPG	CO ₂	Yes	Major emission sources.
		CH ₄	No	Minor emission source.
		N ₂ O	No	Minor emission source.
Project	Emissions caused by the project activity	CO ₂	No	According to methodology, the project is a wind power project, so it does not involve project emission.
		CH ₄	No	According to methodology, the project is a wind power project, so it does not involve project emission.

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	According to methodology, the project is a wind power project, so it does not involve project emission.

Figure 3 presents a flow diagram within the project boundary which physically delineates the project activity and its relevant information.

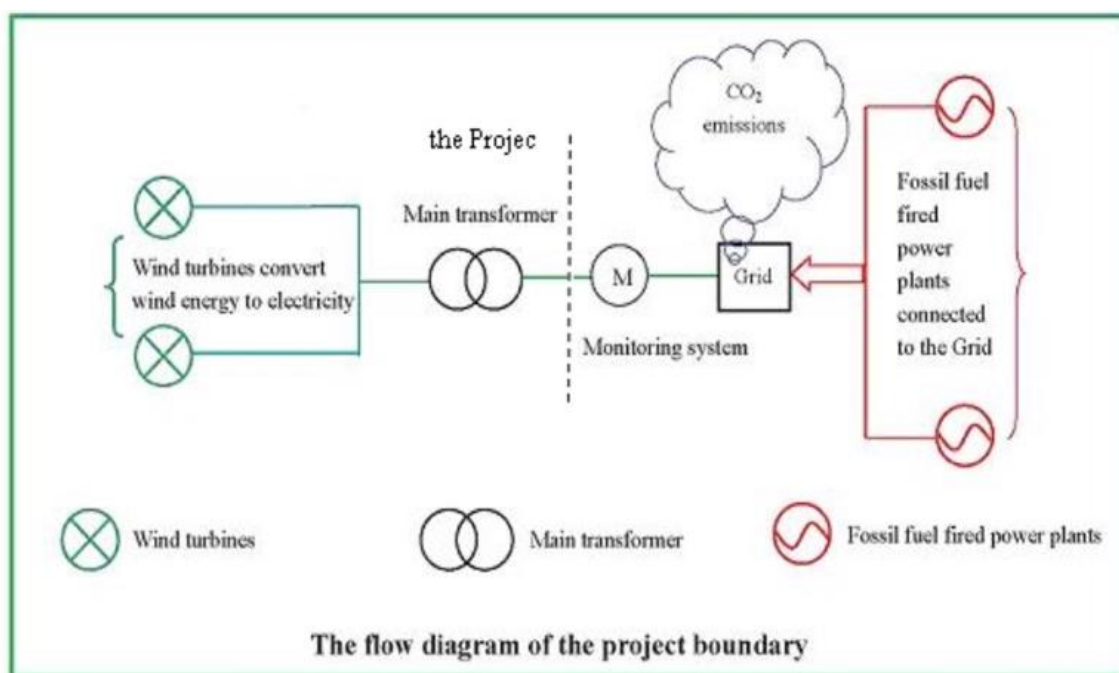


Figure 3 Flow diagram of the project boundary

3.4 Baseline Scenario

The baseline scenario of the project is electricity delivered to NCPG by the project that would otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. According to the methodology ACM0002 (version 20.0), the baseline emissions are the electricity produced by the project multiplied by the emission factor of NCPG.

For the second crediting period, the continued validity of the original baseline should be assessed.

According to the Methodological Tool "Assessment of the validity of the original/current baseline and update the baseline at the renewal of the crediting period" (version 03.0.1), the stepwise procedure as follows should be adopted to assess the continued validity of the baseline and to update the baseline:

Step 1: Assess the validity of the current baseline for the next crediting period

Step 1.1: Assess compliance of the current baseline with relevant mandatory national and/or sectoral policies

There are no new national and/or sectoral policies that could affect the baseline scenario during the renewal of the crediting period. Although national policies favour the development of renewable energy sources, total renewable resources-based power generation accounts for less than 50% of total generation of the NCPG in the latest 5 years respectively (respectively 10.10%, 8.94%, 7.21%, 6.39% and 5.92% in 2017, 2016, 2015, 2014 and 2013)¹. Hence in the absence of the project activity electricity would still have been generated in the existing fossil fuel power plants or by the addition of new fossil fuel power plants connected to the NCPG.

Step 1.2: Assess the impact of circumstances

Firstly, the baseline scenario identified at the validation of the project activity was the continuation of the current practice without any investment;

Secondly, the mainly investment environment or market characteristics especially the feed-in tariff, the policy in terms of market access permit have no significant change which would impact the current baseline. The current practice for the baseline emissions is still the GHG emitted by NCPG: the equivalent electricity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources within the NCPG;

Step 1.3: Assess whether the continuation of the use of current baseline equipment(s) or an investment is the most likely scenario for the crediting period for which the renewal is requested

The current baseline scenario is the continuation of the current practice. In the absence of the project, the electricity would have been supplied by NCPG, and it will not request an investment by the project proponent or third party. So, this step is not applicable.

Step 1.4: Assessment of the validity of the data and parameters Since there are some parameters, which were determined at the start of the first crediting period and not monitored during the first crediting period, are not valid anymore, therefore, the current baseline emissions need to be updated for the second crediting period according to this tool.

Before the time of requesting renewal of the crediting period, the China DNA has issued the latest Notice “2019 Baseline Emission Factors for Regional Power Grids in China” on 29/12/2020², so the emission factor of NCPG is updated for the second crediting period according to this Notice.

In summary, the emission factor of NCPG and all values have been updated to the latest data for the second crediting period according to this Notice.

¹ China Electric Power Yearbook (2014, 2015, 2016, 2017,2018)

² http://www.mee.gov.cn/ywgz/ydqbhb/wsqtkz/202012/t20201229_815386.shtml

Application of Steps 1.1, 1.2, 1.3 and 1.4 confirmed that the current baseline is valid for the second crediting period, but data and parameters need to be updated. Therefore step 2 is used.

Step 2: Update the current baseline and the data and parameters

Step 2.1: Update the current baseline

The baseline emissions for the second crediting period have been updated, without reassessing the baseline scenario, based on the latest approved version of the methodology ACM0002. More details for the updated baseline emissions for the second crediting period can be seen in section 4.

Step 2.2: Update the data and parameters

As mentioned in step 1.4 above, all parameters regarding the grid emission factor calculation have been updated for this second crediting period. More details can be seen in section 4.

3.5 Additionality

According to VCS standard 4.3, a full reassessment of additionality is not required.

The validity of the original baseline scenario is demonstrated in Section 3.4, and it was concluded that the current baseline is still valid for the next crediting period. Only EF related parameters should be updated for the second crediting period.

Project has no change compared with the original status. Therefore, the project description is the same as in the first crediting period.

The demonstration of regulatory surplus is provided below.

The project activity is in compliance with current laws and regulations in China, such as Renewable Energy Law of the People's Republic of China and Environmental Protection Regulation for Wind Power Project, etc. The law and regulations are described in section 1.14. There is no other regulatory requirement for the implementation of a wind power technology.

3.6 Methodology Deviations

There are no methodology deviations for this project.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to the methodology, the baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions (BE_y) is the product of the baseline emissions factor ($EF_{grid,CM,y}$ in tCO₂e/MWh) calculated, times the electricity supplied by the project activity to the grid (EG_y in MWh), as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

Where:

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ / mass or volume unit)

Calculation of $EF_{grid,CM,y}$

Following ACM0002 (version 20.0), the baseline emission factor (EF_y) is calculated as a combined margin ($EF_{grid,CM,y}$), consisting of the combination of operating margin ($EF_{grid,OM,y}$) and build margin ($EF_{grid,BM,y}$) factors according to the following six steps defined in the TOOL 07: “Tool to calculate the emission factor for an electricity system” (version 07.0).

Data for the calculations are based on official data on “2019 Baseline Emission Factors for Regional Power Grids in China” on 29/12/2020.

Step 1. Identify the relevant electricity systems

In accordance with the boundary definitions of the China DNA, the spatial extent of the project boundary includes the Project and all power plants connected physically to the NCPG that the Project is connected to. The China DNA has published a delineation of the project electricity system and connected electricity systems, this delineation is used. Following the DNA delineation, the project electricity system is the North China Power Grid (NCPG), which consists of Beijing, Tianjin, Hebei, Shanxi, Inner Mongolia Autonomous Region, and Shandong Province Power Grids. The project is located in Shandong Province and there’s electricity export to NCPG. Therefore, NCPG is identified as the relevant electric power system.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional)

For this project, Option I (only grid power plants are included in the calculation) is chosen.

Step 3. Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on the following methods:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch Data Analysis OM, or
- (d) Average OM.

As the low-cost / must-run resources constituted less than 50% of total electricity generation of NCPG in recent five years (respectively 10.10%, 8.94%, 7.21%, 6.39 % and 5.92% in 2017, 2016, 2015, 2014 and 2013)³, the Simple OM (a) method is selected and the following data vintage is chosen to calculate the emission factor:

Ex ante option: use a 3-year generation-weighted average, based on the most recent data available, without requirement to monitor and recalculate the emissions factor during the crediting period. And according to the tool, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

Step 4. Calculate the operating margin emission factor according to the selected method

The Simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units. The simple OM may be calculated:

- Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit, or
- Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

- (a) The necessary data for Option A is not available; and
- (b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
- (c) Off-grid power plants are not included in the calculation.

³ China Electric Power Yearbook (2014, 2015, 2016, 2017, 2018)

In this project, all of the above conditions can be met, so Option B was chosen.

Under this option, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

For Option B, the Simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants /units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_{i,m} FC_{i,y} \cdot NCV_{i,y} \cdot EF_{CO2,i,y}}{\sum_m EG_y} \quad (2)$$

Where:

- $EF_{grid,OMsimple,y}$ = Simple operating margin CO₂ emission factor in year y (tCO₂e/MWh)
- $FC_{i,y}$ = Amount of fossil fuel type i consumed in the project electricity system in year y (mass or volume unit)
- $NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ / mass or volume unit)
- $EF_{CO2,i,y}$ = CO₂ emission factor of fossil fuel type i in year y (tCO₂e/GJ)
- EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost / must-run power plants / units, in year y (MWh)
- i = All fossil fuel types combusted in power sources in the project electricity system in year y
- y = The relevant year as per the data vintage chosen in Step 3

On the basis of the data available, the three-year average operating margin emission factor is calculated by the DNA as a full-generation-weighted average of the emission factors:

$$EF_{grid,OM,y} = 0.9419 \text{ tCO}_2\text{e/MWh}$$

Step 5. Calculate the build margin emission factor

As per Section 6.5 of TOOL07 (version 07.0), in terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

(b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

In line with China's Baseline emission factors of regional grids 2019 (BEF₂₀₁₉) published by the Development and Reform Commission of China, Option 1 is chosen for the project; the BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5-units}$) and determine their annual electricity generation ($AEG_{SET_{5-units}}$, in MWh);

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET \geq 20\%$) and determine their annual electricity generation ($AEG_{SET \geq 20\%}$, in MWh);

(c) From $SET_{5-units}$ and $SET \geq 20\%$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. In this case ignore Steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample-CDM}$) the annual electricity generation ($AEG_{SET-sample-CDM}$, in MWh); If the annual electricity generation of that set comprises at least 20% of the annual electricity generation of the proposed project electricity system (i.e. $AEG_{SET-sample-CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample-CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

(e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample-CDM > 10yrs}$).

The BM emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \cdot EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad (3)$$

Where:

$EF_{grid,BM,y}$ = Build margin CO_2 emission factor in year y (tCO_2e/MWh)

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$EF_{EL,m,y}$ = CO_2 emission factor of power unit m in year y (tCO_2e/MWh)

m = Power units included in the build margin

y = Most recent historical year for which power generation data is available

As it is difficult to obtain the detailed data on the power generation, fuel consumption and thermal efficiency of each newly built power unit from public documents, a deviation of TOOL07

(07.0) is adopted following the clarifications⁴ given by the CDM EB concerning the BM emission factor calculation:

- 1) The CDM EB suggested using the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin.
- 2) The EB agreed the use of capacity additions during last 1 ~ 3 years for estimating the build margin emission factor for grid electricity.
- 3) The EB also agreed to use of weights estimated using installed capacity in place of annual electricity generation.

The newly built power plants in the past few years are bundled into “grouped new power plant” according to their construction year, their province and their fuel type. The annual net electricity generation in the year y of each “grouped new power plant” $EG_{m,y}$ is estimated according to their total capacity and the average utilization hours, as the following equation:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad (4)$$

where:

$EG_{m,y}$	=	Annual net electricity generation the unit m in year y (MWh)
CAP_m	=	Installed capacity of the unit m (MW)
$H_{m,y}$	=	Utilization hour of the unit m in the year y (h), determined according to the average utilization hour of the same type of unit in the same province
y	=	The most recent year for which the generation data is available. For the calculation of BM in 2019, $y = 2017$
m	=	grouped new power plant

Since the newly built power plants in the same province (A), in the same year (t) and using the same fuel type (k) are grouped into “a grouped new power plant”, CAP_m represents the total installed capacity of fuel type k power plants located in the provinces A and in the year t :

$$CAP_m = CAP_{A,t,k} \quad (5)$$

where:

⁴ “Request for clarification on use of approved methodology AM0005 for several projects in China”, the EB’s guidance on DNV deviation request.

http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP10ZAK6V5YXPQKK7WYJ

CAP_m	=	Installed capacity of the unit m (MW), with m representing the specified combination of A, t, and k
$CAP_{A,t,k}$	=	Total installed capacity of fuel type k power plants located in the province A and in the year t
A	=	Provinces covered by the NCPG, namely, Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province, Inner Mongolia Autonomous Region.
t	=	Years related to the grouped new power plants, for the 2019 calculation, t represents 2017, 2016, 2015.... Until the aggregated electricity generation of the grouped new power plants reaches 20% of the total electricity generation of the NCPG.
k	=	Fuel type of the grouped new power plants, including hydro, thermal (coal, gas, oil, waste incineration, other thermal), nuclear, wind, solar and others.

The emission factors of each fuel type $EF_{EL,m,y}$ are determined according to the Option A2 in the TOOL07, as the following equation:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad (6)$$

where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
$EF_{CO2,m,i,y}$	=	Average CO ₂ emission factor of fuel type i used in power unit m in year y (t CO ₂ /GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit m in year y (ratio)
m	=	All power units serving the grid in year y except low-cost / must-run power units
3.6	=	Conversion factor (GJ/MWh)

Among the fuel types, the emission factors of hydro, nuclear, wind, solar, other thermal and others are 0. Concerning the emission factors of coal, gas, oil and waste incineration, Equation takes the following form due to conservativeness:

$$EF_{best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{best,y}} \quad (7)$$

where:

$EF_{best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year y (t CO ₂ /MWh)
$\eta_{best,y}$	=	Power generation efficiency of the best technology commercially available in year y
m	=	Power units serving the grid with coal, gas, oil or waste incineration in year y

On the basis of the data available, the three-year average build margin emission factor is calculated by the DNA as a full-generation-weighted average of the emission factors:

$$EF_{grid,BM,y} = 0.4819 \text{ tCO}_2\text{e/MWh}$$

Step 6. Calculate the combined margin emission factor

The calculation of the combined margin emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

- (a) Weighted average CM; or
- (b) Simplified CM.

The weighted average CM method (option A) should be used as the preferred option.

The simplified CM method (option b) can only be used if:

- a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and
- b) The data requirements for the application of step 5 above cannot be met.

The baseline emissions factor (EF_{CM}) is calculated as the weighted average of the Operating Margin emission factor and Build Margin emission factors following ACM0002.

For wind projects, the default weights are as follows: $\omega_{OM} = 0.75$ and $\omega_{BM} = 0.25$:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times \omega_{OM} + EF_{grid,BM,y} \times \omega_{BM} \quad (8)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission for the project electricity system factor in year y (tCO ₂ e/MWh)
$EF_{grid,OM,y}$	=	Operating margin CO ₂ emission factor for the project electricity system in year y (tCO ₂ e/MWh)

ω_{OM} = Weighting of operating margin emissions factor (%)

ω_{BM} = Weighting of build margin emissions factor (%)

On the basis of these weights for the first crediting period, the combined margin emission factor is calculated, and fixed ex-ante:

$$EF_{grid,CM,y} = 0.9419 \times 0.75 + 0.4819 \times 0.25 = 0.8269 \text{ tCO}_2/\text{MWh}$$

Calculation of $EG_{PJ,y}$

The project is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, then:

$$EG_{PJ,y} = EG_{facility} \quad (9)$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

4.2 Project Emissions

According to ACM0002 (version 20.0), project emission of the proposed wind farm project is zero. $PE_y = 0$.

4.3 Leakage

According to ACM0002 (version 20.0), no leakage is considered. The main emissions potentially giving rise to leakage are neglected.

4.4 Net GHG Emission Reductions and Removals

The annual emission reductions ER_y for the project activity are calculated as the baseline emissions minus the project emissions and minus the leakage emissions. Being the project is a zero-emission activity the final GHG emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad (10)$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emissions in year y (tCO₂e/yr)

PE_y = Project emissions in year y (tCO₂e/yr)

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
8/11/2021-7/11/2022	81,885	0	0	81,885
8/11/2022-7/11/2023	81,885	0	0	81,885
8/11/2023-7/11/2024	81,885	0	0	81,885
8/11/2024-7/11/2025	81,885	0	0	81,885
8/11/2025-7/11/2026	81,885	0	0	81,885
8/11/2026-7/11/2027	81,885	0	0	81,885
8/11/2027-7/11/2028	81,885	0	0	81,885
8/11/2028-7/11/2029	81,885	0	0	81,885
8/11/2029-7/11/2030	81,885	0	0	81,885
8/11/2030-7/11/2031	81,885	0	0	81,885
Total	818,850	0	0	818,850

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	“2019 Baseline Emission Factors for Regional Power Grids in China” issued by China DNA
Value applied	0.9419
Justification of choice of data or description of measurement methods and procedures applied	Official statistical data

Purpose of Data Purpose of Data	Calculation of baseline emissions
Comments	/

Data / Parameter	EF _{grid,BM,y}
Data unit	tCO ₂ /MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	“2019 Baseline Emission Factors for Regional Power Grids in China” issued by China DNA
Value applied	0.4819
Justification of choice of data or description of measurement methods and procedures applied	Official statistical data
Purpose of Data Purpose of Data	Calculation of baseline emissions
Comments	/

Data / Parameter	W _{OM}
Data unit	%
Description	Weighting of operating margin emissions factor
Source of data	“Tool to calculate the emission factor for an electricity system” (Version 07.0)
Value applied	75
Justification of choice of data or description of measurement methods and procedures applied	Based on the requirements of “Tool to calculate the emission factor for an electricity system” (Version 07.0)
Purpose of Data Purpose of Data	Calculation of baseline emissions
Comments	/

Data / Parameter	W_{BM}
Data unit	%
Description	Weighting of build margin emissions factor
Source of data	"Tool to calculate the emission factor for an electricity system" (Version 07.0)
Value applied	25
Justification of choice of data or description of measurement methods and procedures applied	Based on the requirements of "Tool to calculate the emission factor for an electricity system" (Version 07.0)
Purpose of Data Purpose of Data	Calculation of baseline emissions
Comments	/

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{output,y}$
Data unit	MWh/yr
Description	Total electricity supplied to the Grid by the Project and Phase II project in the year y.
Source of data	Measured by the electricity meter M1
Description of measurement methods and procedures to be applied	The readings of the electricity meter were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	99,026
Monitoring equipment	M1 is a bidirectional meter. Type: DTZ666 Accuracy class: 0.5S Serial number: 3730001000000580579585 Calibration frequency: at least annually

QA/QC procedures to be applied	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2016).
Purpose of data	Calculation of baseline emissions
Calculation method	NA
Comments	/

Data / Parameter	EG _{another,y}
Data unit	MWh/yr
Description	Electricity generated by the Phase II project in the year y.
Source of data	Measured by the electricity meter M3
Description of measurement methods and procedures to be applied	The readings of the electricity meter were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	0
Monitoring equipment	M3 is a directional meter. Type: DTSD341 Accuracy class: 0.5S Serial number: 11060727990140 Calibration frequency: at least annually
QA/QC procedures to be applied	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2016).
Purpose of data	Calculation of baseline emissions
Calculation method	NA
Comments	/

Data / Parameter	EG _{import,y}
Data unit	MWh/yr
Description	Total electricity imported from the Grid by the project and Phase II project through the 110KV line in the year y.
Source of data	Measured by the electricity meter M1
Description of measurement methods and procedures to be applied	The readings of the electricity meter were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	0
Monitoring equipment	M1 is a bidirectional meter. Type: DZT666 Accuracy class: 0.5S Serial number: 3730001000000580579585 Calibration frequency: at least annually
QA/QC procedures to be applied	The accuracy of meters is 0.5S, which is consistent with Technical administrative code of electric energy metering (DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2016).
Purpose of data	Calculation of baseline emissions
Calculation method	NA
Comments	

Data / Parameter	EG _{im-spare,y}
Data unit	MWh/yr
Description	Electricity imported from the Grid by the project and Phase II Project through a spare 10kV line in the year y.
Source of data	Measured by the electricity meter M4
Description of measurement methods	The local power distributor measured the EG _{im-spare,y} by metering instrument operated by the Grid Company.

and procedures to be applied	
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	0
Monitoring equipment	M4 is a unidirectional meter. Type: DTZY188-Z Accuracy class: 1.0 Serial number: 3730001000000750545938 Calibration frequency: at least annually
QA/QC procedures to be applied	The accuracy of meters is 1.0, which is consistent with Technical administrative code of electric energy metering (DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2016).
Purpose of data	Calculation of baseline emissions
Calculation method	NA
Comments	/

Data / Parameter	$EG_{\text{facility},y}$
Data unit	MWh/yr
Description	Net electricity supplied to the Grid by the proposed project in the year y.
Source of data	Measured by the electricity meter M1, M3 and M4
Description of measurement methods and procedures to be applied	The readings of M1, M3 and M4 were continuously measured and monthly recorded. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	99,026
Monitoring equipment	M1 is a bidirectional meter. Type: DTZ666 Accuracy class: 0.5S

	Serial number: 3730001000000580579585 Calibration frequency: at least annually M3 is a directional meter. Type: DTSD341 Accuracy class: 0.5S Serial number: 11060727990140 Calibration frequency: at least annually M4 is a unidirectional meter. Type: DTZY188-Z Accuracy class: 1.0 Serial number: 3730001000000750545938 Calibration frequency: at least annually
QA/QC procedures to be applied	The accuracy of M1, M3 and M4 is 0.5S, 0.5S and 1.0 respectively, which is consistent with Technical administrative code of electric energy metering (DL/T448-2016). Meters were calibrated as per the requirement of Technical administrative code of electric energy metering (DL/T448-2016).
Purpose of data	Calculation of baseline emissions
Calculation method	The net electricity supplied to the Grid by the proposed project were calculated by the following formula: $EG_{\text{facility},y} = EG_{\text{output},y} - EG_{\text{another},y} - EG_{\text{import},y} - EG_{\text{im-spare},y}$
Comments	

5.3 Monitoring Plan

This Monitoring plan will set out a number of monitoring tasks in order to ensure that all aspects of projected greenhouse gas (GHG) emission reductions for the proposed project are controlled and reported. This requires an ongoing monitoring of the project to ensure performance according to its design and that claimed VCUs are actually achieved.

The proposed project adopts the approved consolidated baseline and monitoring methodology ACM0002 “Consolidated methodology for grid-connected electricity generation from renewable sources” (version 20.0) to determine the emission reductions from the net electricity generation from the wind farm. This plan describes in more detail the process.

1. Monitoring Object

The monitoring is to justify the realistic amount of emission reduction from the CDM project. The monitoring plan will provide credible, accurate, transparent and conservative monitoring data and ensure the real, measurable, long-term GHG emission reduction from this project.

2. Management Structure

China Resources Wind Power (Qingdao)Co., Ltd., the project owner will use this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan. This plan should be modified according to actual conditions and requirements of DOE in order to ensure that the monitoring is credible, transparent, and conservative.

The responsibilities of the project staff are as follow:

The General Manager of the project entity will appoint a VCS project manager. The Operational and Monitoring Manager of the plant, the Financial Chief, and the Technical Chief are responsible for the collection of the data and information required in the monitoring plan.

General Manager: To be responsible for supervising the whole monitoring procedure.

Project Manager: To be responsible for data management and compiling monitoring report.

Operational and monitoring manager: To be responsible for collecting data and do internal audit.

Financial chief: To be responsible for collection of sales receipts.

Technical chief: To be responsible for preparing operational reports of the project activity, recording the daily operation of the wind farm, including operating periods, equipment defects, etc.

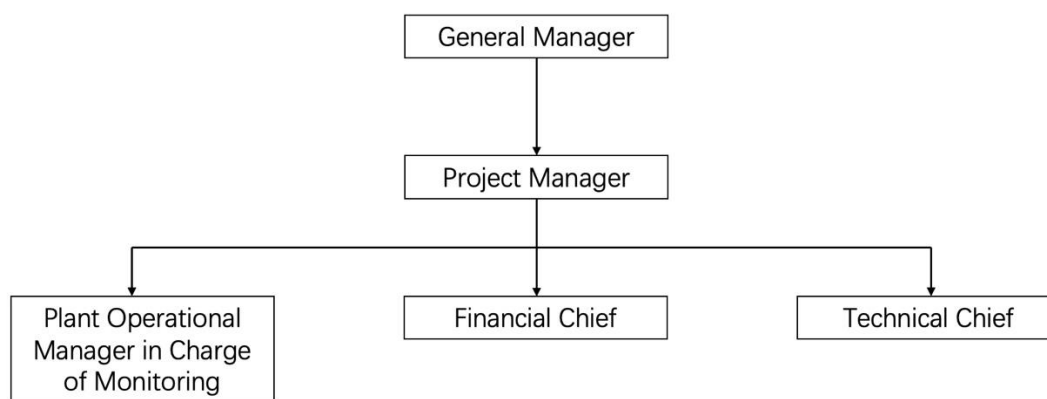


Figure 4 Management Structure of Monitoring Plan

3. Monitoring Program and Equipments

The electricity supplied by the proposed project to the Grid will be transmitted from the on-site substation of the proposed project through the 110kV transmission line and then be delivered to

the Grid and share the 110kV transmission line with the Phase II project, which is developed by the same project owner with the proposed project.

Meter M1 installed on the Tangtian Substation as the main meter will be used to measure the electricity supplied by the proposed project and Phase II project. Meter M2 and M3 installed at the on-site substation will be used to measure the electricity supplied by the proposed project and Phase II project respectively. Sales receipts on the electricity supplied to the grid by Phase II Project will be issued according to meter reading from M3. Sales receipts on the electricity supplied to the grid by the project activity will be issued according to meter reading from M1 minus meter reading from M3. The reading of meter (M2) is carried out by the project owner for cross check. The precision of the meter M1, M2 and M3 will be not lower than 0.5s.

Additionally, a spare 10 kV line is used for supplying electricity in order to start-up or emergencies. Electricity imported from the grid by the project via this spare 10kV line can be measured by a meter (M4) which is owned by the project owner. Reading recording, operation and maintenance of this meter is implemented by the grid. After electricity supplied to the project by the grid through this spare 10 kV line has been confirmed by both sides (the local power distributor and the project owner), electricity sales receipts will be issued accordingly in an approach agreed from both sides and based on the meter reading. The precision of this meter is no lower than 2.0.

The Grid Company will be responsible for the operation of meter M1; the Project Owner will be responsible for the meter M2 and M3. The metering equipments will be annually calibrated and checked for accuracy by the third party designated according to relevant national calibration standard. An agreement will be signed between the proposed project owner and the Grid that defines the metering arrangements and the required quality control procedures to ensure accuracy. The calibration records will be provided to the proposed project owner and maintained by the proposed project owner and the third party designated.

The location of the meters are as the following chart.

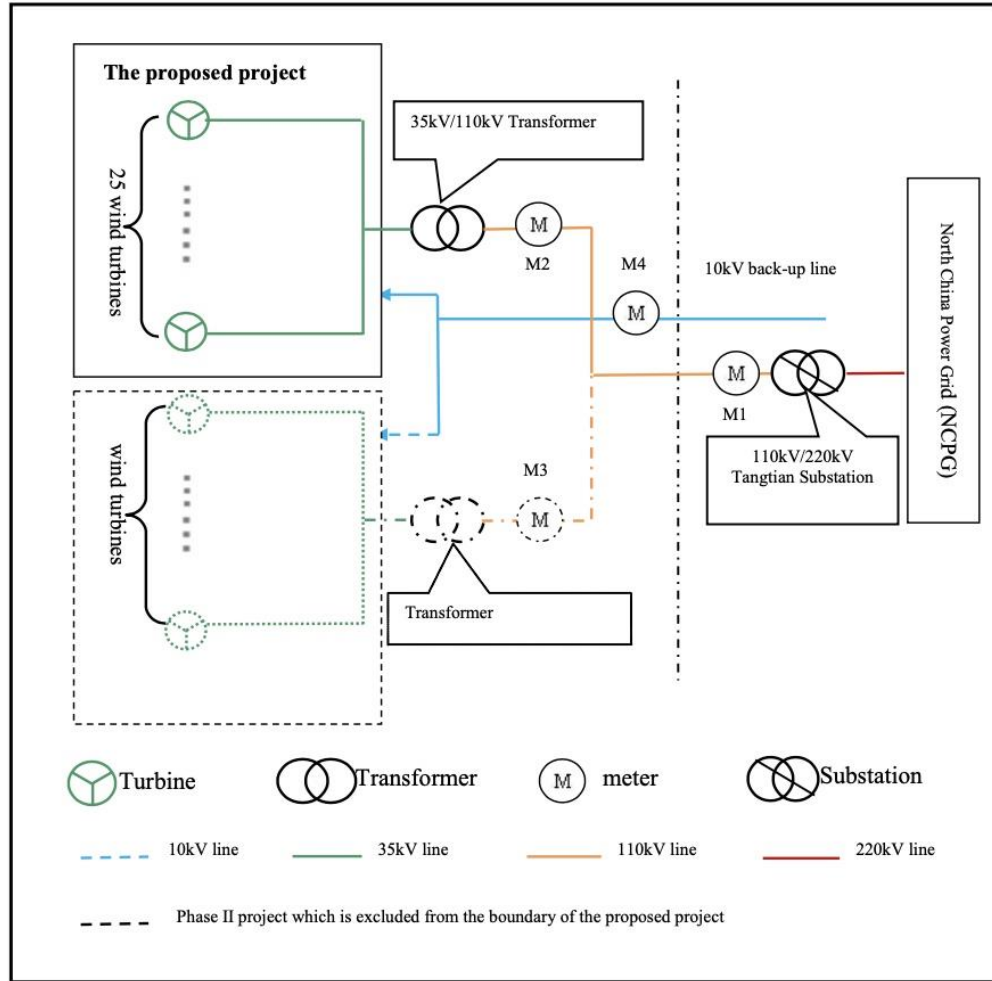


Figure 5 Location of the meters

The net electricity supplied to the grid by the proposed project is calculated as followed:

$$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im-spare,y} \quad (11)$$

Where:

$EG_{facility,y}$ = Net electricity supplied to the Grid by the proposed project in the year y;

$EG_{output,y}$ = Total electricity supplied to the Grid by the proposed project and Phase II project in the year y, which will be measured by bidirectional meter M1 installed in the Tangtian substation

$EG_{import,y}$ = Total electricity imported from the Grid by the proposed project and Phase II project in the year y, which will be measured by bidirectional meter M1 installed in Tangtian substation;

$EG_{im-spare,y}$ = Electricity imported from the Grid through a spare 10 kV line in the year y, which will be measured by meter M4 installed on the spare power line on the site of the proposed project

$EG_{anpther,y}$ = Electricity supplied to the Grid by the Phase II Project in the year y, which will be measured by electricity meter M3;

$EG_{import,y}$ is the total electricity imported from the Grid by the proposed project and the Phase II project in year y, to be conservative, when the $EG_{facility,y}$ is calculated, $EG_{import,y}$ is fully deducted including the electricity imported from the Grid by Phase II project.

In conclusion, the determination of $EG_{facility,y}$ is based on the measurement of electricity meters installed and is conservative.

Before the proposed project is in operation, both the project owner and the grid company will check the equipments to ensure their work properly.

4. Monitoring Training

Monitoring training is critical to ensure that all members of the Monitoring Team have a thorough understanding of the monitoring procedure and are able to carry out the monitoring tasks strictly in line with the requirements. The team leader is responsible for evaluating training outcome. Only qualified staffs can work on duty. The training will include:

Training on operation and monitoring system of the wind farm

This is the type of training which are routinely carried out by the wind farm itself for new staffs.

Training on VCS basics with focus on monitoring

The monitoring manual will be used as the primary training materials. The training includes the following contents:

- VCS project cycle and the significance of monitoring
- Management structure and work scope of each team member
- Components of the monitoring plan
- QA/QC procedure
- Monitoring report template
- Preparation for verification

5. Data Collection

The verification will use the main meter's data as long as the inaccuracy of the meter is within the permissible tolerance. The main procedures are as follows:

I According to the requirements of power purchase/sales agreement, the project owner and the grid company should collect the two meters' data monthly, and check them at the same time.

II The project owner supplies the electricity to the grid company, and provides an electricity sales receipt to the Grid Company. A copy of the sales receipts is stored by the project owner, together with a record of the payment by the grid company.

III When the electricity generated by this project cannot meet the electricity requirement of the power plant, the grid company supplies the electricity to the project owner. The Grid Company provides an electricity sales receipt to the project owner and the sales receipts is stored by the project owner.

IV The project owner records the power supplied to and purchased from the grid, and hence calculate the net electricity supplied to the grid;

V The project owner keeps and safe keeps the records of the main meter's data readings for verification by the DOE.

In case metering equipment is damaged and no reliable readings can be recorded the project entity will estimate net supply by the proposed project activity according to the following procedure:

In case metering equipment is damaged: The project entity and the grid company will jointly calculate a conservative estimate of power supplied to the grid. A statement will be prepared indicating

I the background to the damage to metering equipment;

II the assumptions used to estimate net supply to the grid for the days for which no record could be recorded;

III the estimation of power supplied to the grid. The statement will be signed by both a representative of the project entity as well as a representative of the grid company.

The project entity will furthermore document all efforts taken to restore normal monitoring procedures.

6. Calibration of Meters & Metering

The metering equipment will be properly calibrated and checked according to the rules of Relative Technical Administrative Code of Electric Energy Metering (DL/T448-2016). The project owner will prepare backup procedures to deal with any errors occurred to the meters. The calibration

records carried out by the grid company should be provided to the proposed project owner, and these records will be maintained by the proposed project owner and the third party designated.

Meters should be tested by a qualified metric organization co-authorized by the project owner and the grid company within 10 days after:

I The detection of the reading difference between the main meter and the backup meter that exceeds the allowable tolerance.

II The equipments malfunction caused by improper operation

All the calibration test records should be maintained safely for the verification.

7. Data Management System

To keep safely the record of the data collected during monitoring, this project will set up a complete data management system. The project will perfect the whole monitoring procedure by developing the manual, tracking information from the primary source to the end-data calculations in paper document format. It is the responsibility of the proposed project owner to provide additional necessary data and information for validation and verification requirements of respective DOE. Physical documentation such as paper-based maps, diagrams and environmental assessment will be collated in a central place, together with this monitoring plan. All paper-based information will be stored by the proposed project owner and kept at least one copy.

At the end of each month, the monitoring data will be filed in a spreadsheet and stored on a hard disk, and the paper-based printout should be also archived. Furthermore, the project owner collects the sales receipts for the electricity supplied to the grid as a cross-check, and compiled the monitoring report including the monitoring data and relevant evidence at the end of each crediting year.

All the data will be kept for two years following the end of the last crediting period.

8. Monitoring Report

After the project manager collects and sorts the monitored data, the monitoring report is prepared by the project developer. The project developer has to make sure that the format and content of the monitoring report are consistent with the monitoring methodology in the registered PD.

APPENDIX